LEVEL UP ACADEMY

Class 10 | GSEB | Science

Ch 1 Test (GSEB pattern): Chemical Reactions and Equations

Chapter / Topic: Chemical Reactions and Equations

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

SECTION A

9 marks

Choose the correct option.

Q1. Which of the following is a correctly balanced chemical equation? **[1]**

(a) $Fe + H_2O \rightarrow Fe_3O_4 + H_2$
(b) $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
(c) $3Fe + 2H_2O \rightarrow Fe_3O_4 + 2H_2$
(d) $Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

Q2. Which of the following is NOT a chemical change? **[1]**

(a) Melting of ice
(b) Rusting of iron
(c) Digestion of food
(d) Burning of paper

Q3. When dilute hydrochloric acid is added to zinc granules, the gas evolved is: **[1]**

(a) oxygen
(b) chlorine
(c) carbon dioxide
(d) hydrogen

Q4. In the reaction $2H_2S + SO_2 \rightarrow 3S + 2H_2O$, the substance that is oxidised is: **[1]**

(a) SO_2
(b) S
(c) H_2S
(d) H_2O

Q5. $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ takes place in sunlight. This is an example of: **[1]**

(a) photolytic decomposition
(b) combination reaction
(c) thermal decomposition
(d) displacement reaction

Q6. Which of these is an endothermic process? **[1]**

(a) Burning of natural gas
(b) Respiration
(c) Decomposition of calcium carbonate
(d) Reaction of quicklime with water

Q7. In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, the substance that gets oxidised is: **[1]**

(a) CuO
(b) H_2
(c) Cu
(d) H_2O

Q8. When lead nitrate is heated strongly, the products are: **[1]**

(a) lead and nitrogen
(b) lead oxide and nitrogen
(c) lead nitrite and oxygen
(d) lead oxide, nitrogen dioxide and oxygen

Q9. Which of the following is an exothermic process? **[1]**

(a) Electrolysis of water
(b) Respiration

(c) Decomposition of calcium carbonate
(d) Photosynthesis

SECTION B

6 marks

Q10. Why should a magnesium ribbon be cleaned with sandpaper before burning it in air? Write the balanced equation for its burning. **[2]**

Q11. What is an oxidation reaction? Identify the substance oxidised in: $4Na + O_2 \rightarrow 2Na_2O$. **[2]**

Q12. What is a precipitation reaction? Give one example with a balanced chemical equation. **[2]**

SECTION C

6 marks

Q13. A solution of a substance 'X' is used for whitewashing walls. **[3]**

(i) Name 'X' and write its chemical formula. **[1]**

(ii) Write the balanced equation for the reaction of X with water. **[1]**

(iii) Why does the wall get a shiny finish two to three days after whitewashing? **[1]**

Q14. A shiny brown element X becomes black when heated in air. **[3]**

(i) Identify X and the black substance. **[1]**

(ii) Write the balanced equation. **[1]**

(iii) Name the type of reaction. **[1]**

SECTION D

4 marks

Read the passage and answer the questions that follow.

Q15. In a class activity, students heated lead nitrate powder in a boiling tube. Brown fumes were given off and a yellowish residue was left behind. In another observation, a copper vessel kept in the open for a few months developed a green coating on its surface. **[4]**

(i) Name the brown gas given off when lead nitrate is heated. **[1]**

(ii) What type of reaction is the heating of lead nitrate? **[1]**

(iii) What is the green coating on the copper vessel, and why does it form? **[2]**

~ All the best! ~